

An improved method for cloth pattern cutting based on Holistically-nested Edge Detection

Naigong Yu^{1,2,3}, Zhen Zhang^{1,2,3}, Qiao Xu^{1,2,3}, Essaf Firdaous^{1,2,3}, Jia Lin^{1,2,3},

¹ Beijing University of Technology, Faculty of Information Technology, Beijing, China, 100124

² Beijing Key Laboratory of Computing Intelligence and Intelligent System, Beijing, China, 100124

³ Engineering Research Center of Digital Community, Ministry of Education, Beijing, China, 100124
E-mail: yunaigong@bjut.edu.cn

Abstract: Image edge detection is the basis for precise positioning and accurate cutting of cloth pattern contours. Compared with the commonly used traditional edge detection methods, the Holistically-nested Edge Detection has clearer and more continuous detection results including the reduction of the false detection rate. However, this method extracts a coarser thick outline. In order to extract a high-precision cloth pattern outline, clearly distinguish the main body of the pattern from the background, provide convenience for the follow-up cutting machine for accurate cutting, this paper proposes an improved method for edge detection of cloth pattern cutting based on the holistically-nested Edge Detection method. The edge refinement and smoothing process are added, where the edge detection, edge refinement, and edge smoothing of the clothes images are carried out in sequences, so that the extracted cloth pattern contour is continuous, smooth, and detailed, allowing the respect of the cutting requirements of the cutting machine and the requirements of the factory production.

Key Words: Holistically-nested edge detection, edge detection, cloth pattern contour, edge refinement and edge smoothing

1 Introduction

With the rapid development of industrial modernization, people have higher and higher requirements for the quality of various fabric patterns. Edge detection is an important part of the cloth patterns production process, and the result directly affects the accuracy of subsequent cutting. How to accurately identify the edge of the main body contour and ensure the continuity and smoothness of the edge have become the focus of edge detection research.

There are many classic edge detection methods such as Sobel operator, Robert operator, Canny operator, and others [1]. Among them, the Sobel operator uses the discrete difference between rows and columns and uses filters to convolve the image. This method is not suitable for complex image noise. In Robert's method, the vertical and horizontal edges are identified separately and then combined later. This method usually produces a wider response in the area near the edge of the image, the accuracy of edge positioning is not very high, and its anti-noise ability is not strong. Canny operator is the optimal algorithm in traditional edge detection algorithms, but its non-maximum suppression process is prone to random errors, resulting in false edges [2, 3, 4]. In summary, traditional edge detection methods mainly use filters to perform convolution operations to achieve edge detection. They are not advanced enough to deal with noise problems. They have large limitations as the difficulty to distinguish between the background and the subject. It is hard to achieve a good edge detection effect on complex scene conditions such as the background.

In recent years, with the increase of profound and in-depth research in the field of image processing, there have been many breakthroughs in edge detection methods.

Mainly due to the huge trend of deep learning and its fascinating results for complex problems, new edge

detection methods based on deep learning have been proposed. For example, the literature [5] mentioned the edge detection algorithm based on spectral clustering, the cross-layer fusion edge detection algorithm, and the multi-scale fusion edges detection algorithm. Among them, Deep Contour is a representative algorithm based on spectral clustering edge detection. In order to enable CNN to learn discriminative features, the algorithm classifies different contour features and uses different model parameters to represent them. Although the idea is simple and easy to implement, it lacks adaptability and relies on experience [5]. The Richer Convolutional Features algorithm in the edge detection algorithm based on cross-layer fusion can increase the receptive field of the feature map and improve its resolution, but this method has the problem of easy loss of edge information. This is not allowed in the cloth pattern cutting process based on edge detection. Holistically-nested edge detection method based on multi-scale fusion uses a fully convolutional neural network and a deep-supervised deep learning model for image to image prediction. The label image used in the training of the network shields the unimportant background edge in the natural image [6, 7], which is of great significance for the cutting of cloth patterns with practical application background. In the process of mass production and transportation, the fabric to be cut may be contaminated, and the holistically-nested edge detection method can be used to shield the contaminated background to a certain extent to reduce edge false detection.

Considering all the Cloth pattern edge detection problems and requirements, this paper uses the holistically-nested edge detection method instead of the traditional edge detection algorithm for edge detection of the cloth pattern to be cut. However, the original holistically-nested edge detection method detects thicker edge lines, which fails to meet the cutting requirements of the cutting machine. So, in order to meet the actual production requirements, we introduced edge refinement and smoothing processes based on the original holistically-nested edge detection algorithm. After using the edge detection method to obtain the

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preliminary image edge, we perform image refinement on the edges of the binary image, and then smooth the edge image, so that the edge contour corresponds to the requirements and is ready to be used by the cutting machine.

2 Method

2.1 Holistically-nested Edge Detection

The holistically-nested edge detection method (HED) mainly has two aspects: training and prediction based on the whole image, and multi-scale and multi-level feature learning. For the first aspect, the HED network refers to the complete convolutional neural network, takes the entire

image as input, and directly generates the edge image as the output. For the second aspect, HED modifies VGG16 [6, 8], by removing the last convolutional layer and the fully connected layer, and add the in-depth supervision [9, 10]. They also insert the side output layers after the convolutional layers. The deeper the network layer, the larger the convolution kernel and the smaller the side output. The final output is a weighted fusion of multiple side output features [6, 11]. Fig 1 shows the structural model of the holistically-nested edge detection method based on the VGG network.

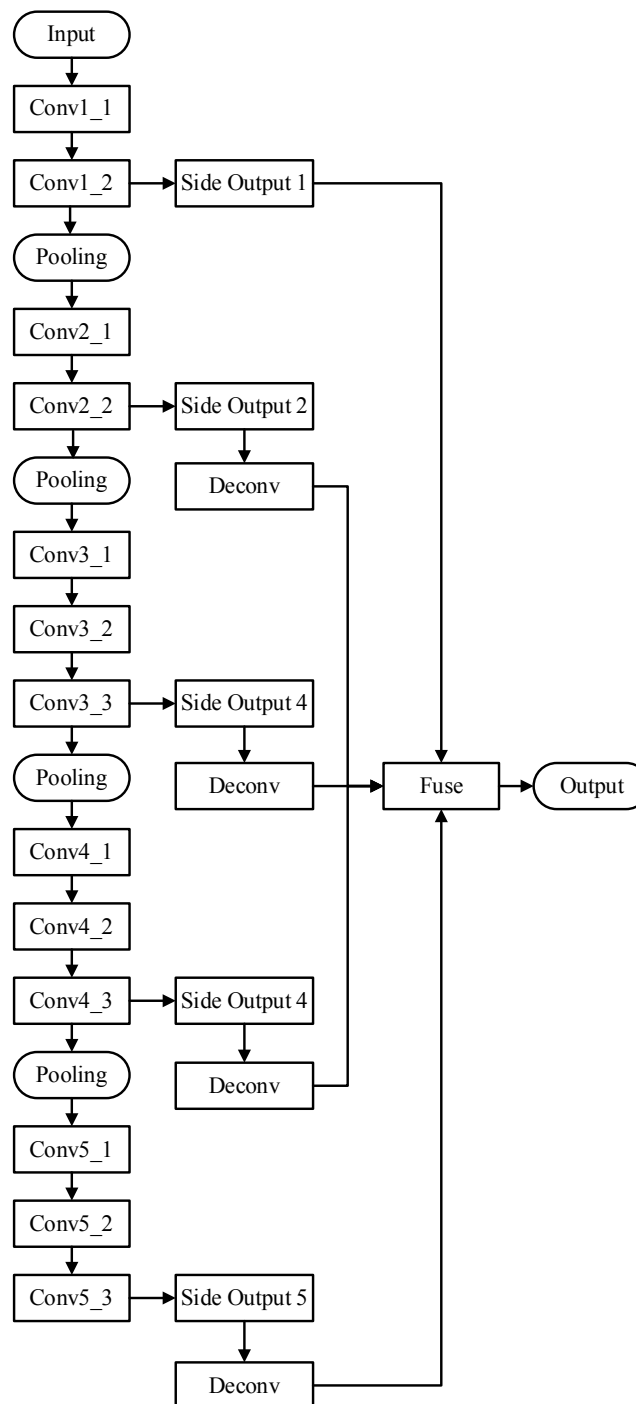


Fig. 1: Holistically-nested Edge Detection

In this network model, we use W to represent the parameter set of all standard network layers, w to represent the corresponding weight of the side output layer, and m to represent the number of layers of the side output layer, and define the loss function of the side output layer, as shown in formula (1), where the function of α_m is to control the proportion of the loss function of each independent side output layer.

$$L_{side}(W, w) = \sum_{m=1}^M \alpha_m l_{side}^{(m)}(W, w^m) \quad (1)$$

The label used in the training of the holistically-nested edge detection model deliberately ignores part of the background, which makes the separation between the edge subject and the background detected by this method clearer. In contrast, traditional edge detection methods such as the Canny operator mainly consider gradient characteristics and use gray-scale changes to detect edges. This often detects all the edges of the background outside the contour of the subject. This is also the main reason why this paper uses the holistically-nested edge detection algorithm to detect the edge for the cloth pattern cutting. In order to directly use the side output layer prediction, a "weighted-fusion" layer is added to the network, thereby connecting the prediction results of all the side output layers [6, 12], and learning the mixed weights in training. The loss function of the fusion layer is shown in formula (2):

$$L_{fuse}(W, w, h) = \text{Dist}(Y, \hat{Y}_{fuse}) \quad (2)$$

$\text{Dist}(\dots)$ represents the distance between the prediction and the actual label after fusion. h represents the fusion weight. And $Y = \{y_j, j = 1, \dots, |X|\}, \{y_j \in \{0,1\}\}$ represent the actual edge map corresponding to the input image, and $\hat{Y}_{fuse}$ represent the edge prediction of the weighted-fusion layer.

The loss function of the entire network is the cumulative sum of the side output loss function and the fusion layer loss function [6], which is presented in formula (3):

$$L = L_{side}(W, w) + L_{fuse}(W, w, h) \quad (3)$$

The objective function is the minimum loss function.

For the cloth edge-cutting process with actual production requirements, the cloth may produce certain contamination or dirt during the transmission process, and redundant pattern points may also appear during the manufacturing process. The holistically-nested edge detection algorithm can more accurately identify the edge of the main pattern, and ignore the unobvious and unimportant parts, which can effectively reduce the false detection rate of the pattern edge.

In addition to effectively reducing the false edge detection rate, this algorithm is also clearer and more accurate than the cloth contour edges detected by traditional edge operators such as Canny. BSDS500 is a classic data set for edge detection [6, 14]. Literature [6] evaluated the parameters of the edge detection task results of the holistically-nested edge detection algorithm and Canny algorithm on BSDS500, and proved that the edge detection effect of the holistically-nested edge detection algorithm is significantly better than Canny's.

2.2 Edge Contour Refinement

For the detected edge image, the edge contour is clear, and it has a certain shielding effect on the unimportant background. However, the edge lines are relatively thick. For simple edge detection, thicker line contours can meet the requirements. However, in the production process of fabric pattern cutting, precise positioning of the edge lines is required to facilitate accurate cutting. Therefore, it is necessary to refine the contour of the existing edge image.

Refinement refers to removing some useless points from the original image according to certain principles, and ensuring that the image can maintain its original shape [14]. The refinement method of non-maximum suppression proposed in [6] is prone to false edges, which is not allowed for edge detection of patterns to be cut with the actual operating background. This paper adopts a binary image thinning algorithm based on the index table. Its specific principle is as follows: corresponding to the index table, judge whether each edge point can be deleted according to the different conditions of the eight connected domains of each edge point. Assuming that the edge point is P, The eight-connected domain refers to the eight points up, down, left, right, top left, bottom left, top right, and bottom right around point P, as shown in Table 1. Its specific criteria are: internal points cannot be deleted; encouraged points cannot be deleted; straight endpoints cannot be deleted; for boundary points, if the connected component does not increase after removal, the boundary point can be deleted [15].

Table 1: Schematic diagram of eight connected domains

P0	P1	P2
P7	P	P3
P6	P5	P4

According to the above principle, for a specified edge point P of a binary edge image with a white background and black edges, the eight points in the eight neighborhoods are respectively corresponding to a multiplier, P0 corresponds to multiplier 1, P1 corresponds to multiplier 2, P2 corresponds to multiplier 4, P3 corresponds to multiplier 16, P4 corresponds to multiplier 128, P5 corresponds to multiplier 64, P6 corresponds to multiplier 32, and P7 corresponds to multiplier 8. Set the weight of the black pixel to 0 and the weight of the white pixel to 1, and add the multiplier of the white position to get the total value of the edge point. Then look up the index table, determine whether the corresponding item of the position of the total value is to be retained, if the corresponding item of the index table is 0, keep it, otherwise remove it. In this way, the original edge line is thinned, so that the position of the edge to be cut can be accurately located [16, 17].

2.3 Smooth Edges

The resulting refined edge by the above method is not smooth enough. If the cutting is directly performed according to the edge coordinate data at this time, the cutting will not be ideal, and the production requirements cannot be met. Therefore, smoothing processing for the refined edges is necessary. Considering the actual cutting requirements, we use a median filter algorithm with better contour

protection to smooth the edges [17, 19], and try to smooth the contour as much as possible while ensuring to keep the definition and accuracy of the contour. The formula for the median filtering algorithm is shown in formula (4):

$$\bar{f}(n) = \text{med}[x(n-N)...x(n)...x(n+N)] \quad (4)$$

where $\bar{f}(n)$ represents the gray value of the center pixel after filtering and smoothing, $x(n-N)...x(n)...x(n+N)$ represents all pixels in the mask, and $\text{med}[x(n-N)...x(n)...x(n+N)]$ represents the median operation after all pixels in the mask are arranged in ascending or descending order. Smoothing is performed by setting the gray value of the pixel to the median value of the gray value of all pixels in a certain neighborhood window of that point [20].

3 Experimental Results and Analysis

The experiments were done on a dual Graphics Processing Unit (GPU) mode. The used server characteristics are as follows: an Ubuntu 16.04 64 bits as an operating system, with a memory of 16 GB, with a processor clock 33 MHz, and Graphics of Tesla K40c.

The original HED algorithm uses the BSDS dataset, with 200 pictures for training and 200 pictures for testing. In this paper, under the TensorFlow framework, and using data augmentation, we used 1500 expanded images obtained by rotating the BSDS500 data set at different angles to train the holistically-nested edge detection model. After 5000 iterations, the loss function reaches stability and convergence. Actual production edge cutting requires accurate and smooth edges. The average value, standard deviation, and average gradient are commonly used

indicators to verify image quality, in addition to the consideration of the practical application background of this method. Therefore, this paper adopts a method of combining subjective evaluation and objective evaluation for the image edge detection algorithm, and evaluates the detection effect from two aspects: actual production cutting requirements and image quality evaluation indicators.

First, we use the holistically-nested edge detection algorithm and the Canny algorithm to perform edge detection on the same pattern respectively. The pattern used here is the cloth pattern commonly used in the actual cutting operation of the factory. Fig 2 shows the comparison of the two methods without unnecessary background interference. It can be seen that the edge detected by the holistically-nested edge detection method is significantly better than the Canny operator in terms of clarity and continuity, which is more in line with the actual cutting requirements of the cutting machine.

Then we use two algorithms to detect the cloth's edge with contamination in the background. The comparison result is shown in Fig 3. It can be seen that for non-essential backgrounds such as stains, the holistically-nested edge detection method can better shield and ignore them. But, for Canny operator, it performs edge detection on the background that does not need to be detected, resulting in false detection, which interferes with the subsequent cutting work. From the above results, and in order to correspond and respect the actual cutting operation in the factory, HED has obvious advantages compared with traditional edge detection operators such as Canny. The background false detection rate is lower, and the operation efficiency is improved.

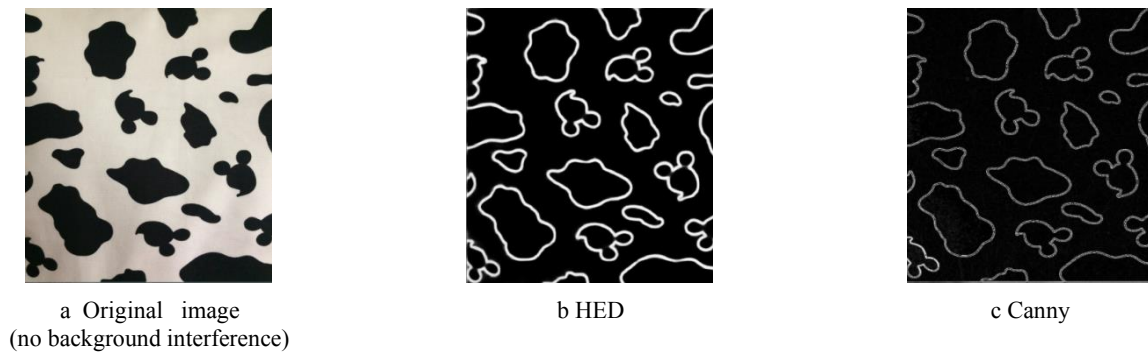


Fig. 2: The comparison of the two methods

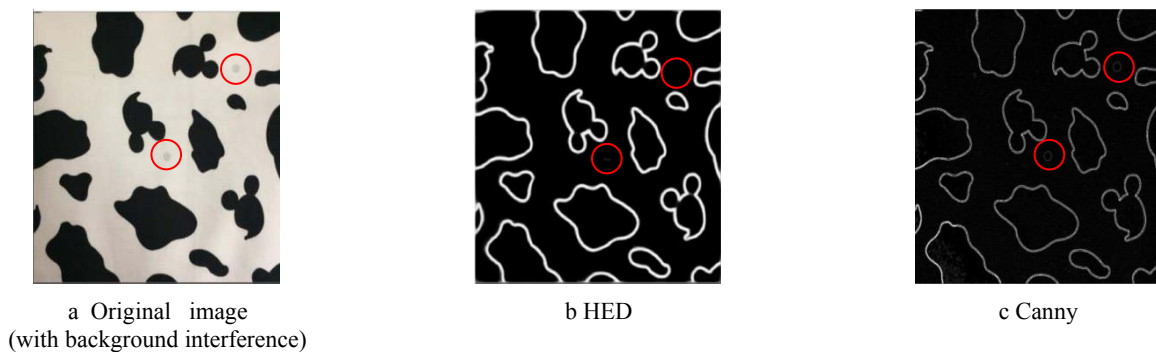


Fig. 3: The comparison of the two methods

Finally, we compared the results of the two methods with three commonly used image quality indicators. The results are shown in Table 2. The average gradient of the image reflects the sharpness of the image. The larger the average gradient, the clearer the image. The mean value is also positively correlated with image brightness, and the standard deviation reflects the degree of dispersion between the image pixel value and the mean value. The larger the standard deviation, the better the image quality. From Table 2, we can see that the image quality obtained using the HED algorithm is significantly better than the Canny algorithm.

Table 2 Image index comparison

Image quality index Method	Image gray average	Image standard deviation	Image average gradient
HED (no background interference)	28.7273	65.0641	20.258
Canny (no background interference)	16.9062	33.5928	11.1169
HED (with background interference)	31.9688	68.5462	18.8969



a HED



b Edge refinement



c Smooth edges

Fig. 4: Improved results

4 Conclusions

Through the comparison of the experimental results, we can know that the holistically-nested edge detection algorithm performs better than the traditional edge detection algorithm in the edge detection of the cloth pattern to be cut with the actual operation background. And it is easier to obtain clear and continuous edge images without background interference. With the addition of refinement and smoothing steps, the improved method makes the edge contour more detailed and smooth, which is more in line with the process requirements of fabric pattern edge cutting than the traditional edge detection.

In recent years, with the rapid development of deep learning, detection algorithms based on CNN have also received widespread attention. The application of related technologies and algorithms to actual engineering operations is of great significance for promoting the development of modern industry in our country.

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interference)			
Canny (with background interference)	13.4369	24.0886	8.8292

Although the use of the holistically-nested edge detection algorithm has obvious advantages, considering the highly accurate positioning requirements for the edge contour in the actual cutting project, we need to refine the edge contour. The effect of using the look-up table method is shown in Fig 4. As shown in Fig 4 b, we can see that compared with the initial edge, the look-up table method has achieved obvious refinement results, and basically meets the cutting requirements of the cutting machine. However, we found that after thinning, the original smooth edge contour is obviously jagged. If you cut directly according to the contour at this time, the cut pattern edge may not be smooth enough, which will easily lead to an unqualified product cut. For this, we use the median filter smoothing algorithm to smooth the edge contour, and the result is shown in Fig 4 c. At this time, the edge contour is continuous, smooth, and detailed, which can meet the edge cutting requirements of the cutting machine.

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